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Related work

Machines and people in mathematics. The closest precedent for the shape of our loop is the use of computation to guide intuition toward statements that people then prove [Davies et al., 2021]; there the direction is machine to human, whereas one leg of our eleventh pivot runs the other way, with the proof supplying a region rather than a candidate. A second line lets search refute conjectures directly, through reinforcement learning over constructions [Wagner, 2021] or program search guided by a language model [Romera-Paredes et al., 2024]; our search is far simpler than either, and it was thorough over a space that excluded the answer. A third line automates proof itself [Trinh et al., 2024, Hubert et al., 2025, The mathlib Community, 2020]; we do none of this, and our proofs are not formalized. Our harness sits instead in the tradition of machine-assisted proof [Appel and Haken, 1977, Hales et al., 2017] and experimental mathematics [Borwein and Bailey, 2008]: it is bookkeeping about how much a numerical verdict is worth.

The substrate. The theory builds on performative prediction [Perdomo et al., 2020, Brown et al., 2022, Jagadeesan et al., 2022, Bracale et al., 2025, Lin and Zrnic, 2024], learning-while-earning [den Boer, 2015, Keskin and Zeevi, 2014, Lai and Robbins, 1979], dual control and least-costly identification [Feldbaum, 1960, Bombois et al., 2006], van Trees bounds [Gill and Levit, 1995], classical and task-optimal design [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006, Wagenmaker et al., 2021, Simchowitz and Foster, 2020], and the REFLEX market [Nagarajan and Ashok, 2026, Guéant et al., 2013, Barzykin et al., 2026].

A Assumptions and formal statements

This section collects the standing assumptions and the formal statements of the results the main text argues from. Identifiers (L1, T2, T9, and so on) are the register identifiers used throughout the supplementary repository, so each statement can be matched to its derivation document, proof, and numerical checks. The register itself, with per-result proof status, is `mlxor-derivations/` `THEOREMS.md`.

A.1 Standing assumptions

A1 (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, with expansions taken at h^* and third-order remainder bounded by $L_3 = \sup |\Phi'''|$. **A2** (exploration class): all policies are non-anticipating; A2-sym adds $\mathbb{E}[u_t | \mathcal{F}_t] = 0$, A2-adaptive is unrestricted, and costs are measured from the certainty-equivalent anchor. **A3** (noise exogeneity): the observation noise ζ_t is i.i.d., mean zero, variance σ^2 , independent of $\mathcal{F}_t$. **A3'** relaxes this to a feedback channel with gain ϕ . **A4** (trust region): $|d_t| \leq r$ pathwise. **A5** (stationary scale): $S_T = \Theta(T)$. **A6** (stable base loop): the retraining modulus satisfies $m < 1$.

The local dynamics are $d_{t+1} = -m d_t + u_t$ with $d_t = h_t - h^*$, observations $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$, and Φ satisfying $\Phi'(h^*) = \delta_0$ and $-\Phi'' = \gamma_{\text{PO}}$ at the operating point. S_{xx} denotes the centered design energy $\sum_{t \leq T} (d_t - \bar{d})^2$.

A.2 The exchange rate and its floors

Lemma 1 (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies $C_T = \frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[\sum_{t \leq T} d_t^2] + R_T$ with $|R_T| \leq \frac{L_3}{6} \mathbb{E} \sum_t |d_t|^3$, and the realized cost’s variance decomposes exactly as $\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}(\sum_t d_t) + \frac{\gamma_{\text{PO}}}{4} \text{Var}(\sum_t d_t^2)$ with vanishing cross term.*

Theorem 1 of the main text is the asymptotic form of the following pathwise identity, which is what is proved in Appendix B.

Theorem 1 (The exchange rate, pathwise; T2). *Under A1, A2-sym, A3 and A6, in the stationary regime, on every realization,*

$$\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2 = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}} \right) = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + O_P(1/T)),$$

independent of the exploration intensity, the horizon, and the modulus.

Theorem 2 (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with $\sum_{t \leq T} d_t^2 \leq D$ almost surely and all estimators, biased included, the minimax risk over an interval of width w satisfies $\inf \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \sigma^2 / (D + \sigma^2(\pi/w)^2)$, so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares. The argument is a van Trees reduction in the style of Gill and Levit [1995].*

Theorem 3 (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator, $\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \gamma_{\text{PO}} \sigma^2 / 27 (1 - O(T^{-1/2}))$, the supremum over the two-point response family at the minimax scale, the constant arising from a Le Cam reduction optimized at $c = 2/3$. The sharp constant $\frac{1}{2}$, against which no simulated policy fell by more than Monte Carlo error, is conjectured and is part of the open problem OPEN-1. It is not proved here or anywhere in the repository.*

A.3 Structure-proofness

Theorem 4 (Structure-proofness of the floor, exact reach; T9). *Fix the structural family $\tau_{\theta}(h) = C_0 + C_1 e^{-ch}$ ($C_1 > 0$, $c > 0$), the anchor $h^* > 0$, the sensitivity $s(h) = \partial_{\theta} \tau_{\theta}(h)$, and the estimand gradient $g = \partial_{\theta} \varepsilon(h^*) = -s'(h^*)$. For a finitely supported design μ define the normalized Cramér–Rao product*

$$R(\mu) = (g^{\top} M(\mu)^{-1} g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^{\top} d\mu,$$

so that $R(\mu) \geq 1$ states that the family-knowing Cramér–Rao product $\text{Var}(\hat{\varepsilon}) \times C_T$ is at least the exchange-rate floor $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$ under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i) Within reach the floor is structure-proof: if $\text{supp}(\mu) \subset [h^ - 2/c, \infty)$, then either $\varepsilon(h^*)$ is not estimable under μ , in which case the bound is infinite, or $R(\mu) \geq 1$, strictly whenever μ places mass off the two equality points $\{h^*, h^* - 2/c\}$, which is every estimable design, since a design supported on those two points alone leaves g outside the range of $M(\mu)$; and $\inf_{\mu} R(\mu) = 1$, approached but not attained by symmetric three-point designs collapsing at h^* . In particular the floor binds on every trust-region design with $r \leq 2/c$. (ii) The reach is exact: the pointwise domination underlying (i) fails strictly below $h^* - 2/c$, and an explicit four-point design with a single below-reach probe of weight 3×10^{-4} attains $R = 0.8517$, computer-verified at 50-digit precision as check V9.2.*

A.4 Other register results

The register also contains T1 (information saturation of the noiseless retraining trajectory), T5a and C5.1 (A-optimal exploration under a value budget and the price of isotropic jitter), T6 (Chebyshev unidentifiability of the curvature c from retraining trajectories), T7 (the anchoring crossover), L4 (perturbed modulus under an estimated response), and P7.1 (safety under pessimism given a confidence sequence). These are stated and proved in the supplementary repository (`mlxor-derivations/latex/proofs.tex` and the corresponding derivation documents) and are not restated here, since the main text does not argue from them; they appear there only as the context that Theorems 1 and 2 of the main text sit in, and in Table S1 as verification rows.

B Proofs

This section proves the two results the main text argues from: the exchange rate of Theorem 1 of the main text, whose exact pathwise form is Theorem 1, and the two-sided structure-proofness result of Theorem 2 of the main text, stated in full as Theorem 4. Both proofs are reproduced from the derivation repository without change; the check identifiers V2.1 and V9.1 to V9.4 refer to its verification suite.

B.1 The cost-equivalence lemma

Proof of Lemma L1 (cost equivalence). Taylor's theorem with Lagrange remainder gives, for each t ,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2} \gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

with $L_3 = \sup |\Phi'''|$ on the operating interval. Summing over $t \leq T$ and taking expectations, it remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = -m d_t + \mathbb{E}[u_t | \mathcal{F}_t] = -m d_t$, so $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0] = 0$ when the loop starts at the operating point or in the stationary regime. The quadratic term contributes $\frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[\sum_t d_t^2]$, which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as $C = -\delta_0 X + \frac{1}{2} \gamma_{\text{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\text{Var}(C) = \delta_0^2 \text{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(Y) - \delta_0 \gamma_{\text{PO}} \text{Cov}(X, Y)$. Under Gaussian innovations the vector $(d_1, \dots, d_T)$ is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\text{Cov}(X, Y) = \sum_{t,s} \mathbb{E}[d_t d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly $-\delta_0 \mathbb{E}[\sum_t d_t]$. $\square$

B.2 The exchange rate

Proof of Theorem T2 (pathwise exchange rate). Under A3, conditionally on the design $\{d_t\}$ the observations are independent with means linear in ε and variance σ^2 , so the least-squares slope has conditional variance $\text{Var}(\hat{\varepsilon} | \text{design}) = \sigma^2 / S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost $\frac{1}{2} \gamma_{\text{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T \bar{d}^2$ gives the displayed identity exactly, on every realization. In the stationary regime $\bar{d} = O_P(T^{-1/2})$ by the stationary long-run variance $\text{Var}(\sum_{t \leq T} d_t) = Tv(1-m)/(1+m)(1+o(1))$, while $S_{xx}/T \rightarrow v$ a.s., so $T \bar{d}^2 / S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

B.3 Structure-proofness, and the exactness of the reach

Proof of Theorem T9 (structure-proofness within reach). Write $x = h - h^*$, $d(h) = x$, and $V = \text{span}\{1, e^{-ch}, h e^{-ch}\}$, the span of the components of s (a bijective correspondence $u \leftrightarrow \phi_u = u^\top s$ between $\mathbb{R}^3$ and V , since $C_1 \neq 0$). Differentiating s gives $s'(h) = (0, -c e^{-ch}, C_1 e^{-ch}(ch - 1))^\top$, so $g = -s'(h^*)$ componentwise, which is the stated estimand gradient.

Step 1: one-dimensional reduction. For $M(\mu) \succ 0$ the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1} g = \sup_{u \neq 0} \frac{(u^\top g)^2}{u^\top M(\mu) u} = \sup_{\phi \in V} \frac{\phi'(h^*)^2}{\int \phi^2 d\mu} = \left(\inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\} \right)^{-1},$$

using $u^\top M u = \int \phi_u^2 d\mu$ and $u^\top g = -\phi'_u(h^*)$ (the sign is lost in the square). Hence

$$R(\mu) = \frac{\int d^2 d\mu}{\inf \{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \}}.$$

Step 2: the candidate and its pointwise domination. Let

$$\phi_0(h) = \frac{2}{c} - e^{-cx} \left(\frac{2}{c} + x \right) = \frac{2}{c} \cdot 1 - e^{ch^*} \left(\frac{2}{c} - h^* \right) e^{-ch} - e^{ch^*} h e^{-ch} \in V,$$

the unique element of V with 2-jet $(0, 1, 0)$ at h^* (the 2-jet map is invertible on V : in the basis $\{1, e^{-ch}, h e^{-ch}\}$ its determinant is $c^2 e^{-2ch^*} \neq 0$); the expansion is $\phi_0(x) = x - (c^2/6)x^3 + O(x^4)$. Writing $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$ we claim

$$|f(x)| \leq |x| \text{ for all } x \geq -2/c, \text{ with equality iff } x \in \{0, -2/c\},$$

and $|f(x)| > |x|$ strictly for every $x < -2/c$. Indeed: (a) $v(x) = f(x) - x$ has $v(0) = 0$ and $v'(x) = e^{-cx}(1 + cx) - 1 < 0$ for $x \neq 0$ (since $1 + t < e^t$ for $t = cx \neq 0$); $v' \leq 0$ with equality

only at the isolated point 0 makes v strictly decreasing on $\mathbb{R}$, so $f > x$ on $x < 0$ and $f < x$ on $x > 0$. (b) $f(0) = 0$ and $f'(x) = e^{-cx}(1 + cx) > 0$ for $x \geq 0$, so $f \geq 0$ on $[0, \infty)$ and, with (a), $0 \leq f \leq x$ there. (c) $q(x) = f(x) + x$ has $q(-2/c) = 0 = q(0)$ and $q'(x) = e^{-cx}(1 + cx) + 1$, where $r(x) = e^{-cx}(1 + cx)$ is strictly increasing on $x < 0$ ($r'(x) = -c^2xe^{-cx} > 0$) from $r(-2/c) = -e^2 < -1$ to $r(0) = 1$; so q' changes sign exactly once on $(-2/c, 0)$, from negative to positive, and $q \leq 0$ there with equality only at the endpoints; with (a) ($f \geq x$, equality only at 0) this gives $x \leq f \leq -x$ on $[-2/c, 0]$. (d) For $x < -2/c$, $r(x) < -e^2$ gives $q' < 0$, so moving left from $-2/c$ the function q increases above 0: $f(x) > -x = |x| > 0$, and in fact $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$ grows exponentially.

Step 3: the bound. If $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$, Step 2 gives $\int \phi_0^2 d\mu \leq \int d^2 d\mu$, so by Step 1, $R(\mu) \geq \int d^2 d\mu / \int \phi_0^2 d\mu \geq 1$. Strictness: equality in Step 2 holds only at $x \in \{0, -2/c\}$, so any μ placing positive mass off $\{h^*, h^* - 2/c\}$ (in particular any with three distinct support points) has $\int \phi_0^2 d\mu < \int d^2 d\mu$ and $R(\mu) > 1$. If $M(\mu)$ is singular, two cases: $g \notin \text{range } M(\mu)$ makes $\varepsilon(h^*)$ non-estimable (infinite bound); $g \in \text{range } M(\mu)$, which does occur for special two-point designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked to obey the bound, e.g. $R = 1.063$ at the pair $\{1.627, 2.127\}$), keeps the sup-representation valid over $\{u : u^\top M u > 0\}$, and $\int \phi_0^2 d\mu = 0$ would put $u_0 \in \text{null } M$ with $u_0^\top g = -1 \neq 0$, contradicting $g \in \text{range } M$; so $\int \phi_0^2 d\mu > 0$ and the bound goes through verbatim. Throughout, the Cramér–Rao functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax sense.

Step 4: the infimum is 1. The Cramér–Rao functional is invariant under C^1 reparametrization, and the 2-jet map $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau_\theta''(h^*))$ is a local diffeomorphism: its Jacobian $[s(h^*) - s'(h^*) \ s''(h^*)]$ has determinant $C_1 c^2 e^{-2ch^*} \neq 0$ (direct computation; V9.4). In jet coordinates η the model reads $\tau = \eta_1 - \eta_2 x + \frac{1}{2}\eta_3 x^2 + O(x^3)$ with sensitivity $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$, and the estimand is η_2 . For the symmetric three-point design μ_δ uniform on $\{h^* - \delta, h^*, h^* + \delta\}$, scale by $D = \text{diag}(1, \delta, \delta^2)$: $M_\eta(\mu_\delta) = D \hat{H}_\delta D$ with $\hat{H}_\delta \rightarrow \hat{H}_0$ invertible (the exact quadratic-model moment matrix), so $(M_\eta^{-1})_{22} = \delta^{-2}(\hat{H}_\delta^{-1})_{22}$ and $R(\mu_\delta) \rightarrow R_{\text{quad}}$. In the exact quadratic model, for any symmetric design (odd moments $m_1 = m_3 = 0$), the $(2, 2)$ cofactor gives $(M_\eta^{-1})_{22} = 1/m_2$ and $R_{\text{quad}} = m_2 \cdot (1/m_2) = 1$. Hence $R(\mu_\delta) \rightarrow 1$ (measured rate $R(\mu_\delta) - 1 = O(\delta^2)$; V9.2), the infimum is 1, and by Step 3 it is not attained.

Step 5: the reach is exact. Below $h^* - 2/c$ the domination of Step 2 fails strictly, and the failure is realized by a design, not only by the candidate: the frozen four-point witness (support $\{1.8726, 1.8665, 1.3073, 0.0500\}$, weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$ after normalization, in the register market $c = 1.5$, frozen anchor $h^* = 1.8461$) has $R = 0.851650 \dots$, verified at 50-digit precision (mpmath) and reproduced in float64 by `run_open1.py` (V9.2). All its support except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor. Symmetric pair-plus-far-probe families do not break it (their ratio approaches 1 from above as the probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches missed it. Verified: V9.1–V9.4. $\square$

C Verification rows

Table S1 lists selected rows of the preregistered harness. Section D plots the rows the main text refers to; the exchange-rate and false-verdict plots are Figure 1 of the main text.

Table S1: Selected rows from the preregistered check harness. The full table (36 rows, 0 FAIL), all seeds, tolerances, and code are in the supplementary repository. The drift row is a deliberately out-of-scope cell, measured and reported rather than excused.

Claim (register)	Measured	Predicted	Verdict
Exchange rate, $m = 0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m = 0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	+16% over rate	out of scope	drift
Feedback floor, $m = 0.3/0.6/0.85$ (T1 corollary)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1 - m^2)$	pass
Design arm $\text{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; $7.8\times$	Fisher 0.248; ≥ 3	pass
Certified agent reaches h_{PO} ; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, $d = 8$ (T5a)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Second economy, 4 cells (T2, exact in LQ); agent	0.384, 0.192; 12.65	0.384, 0.192; 12.5	pass
Multi-book shaped/isotropic steady-state risk (T5a)	1.20 (flat control 1.00)	$> 1.15 (\approx 1)$	pass
Real-data calibration port, 10 cells	10/10 reproduce	within 0.8%	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	≥ 1 ; < 1	pass

D Figures

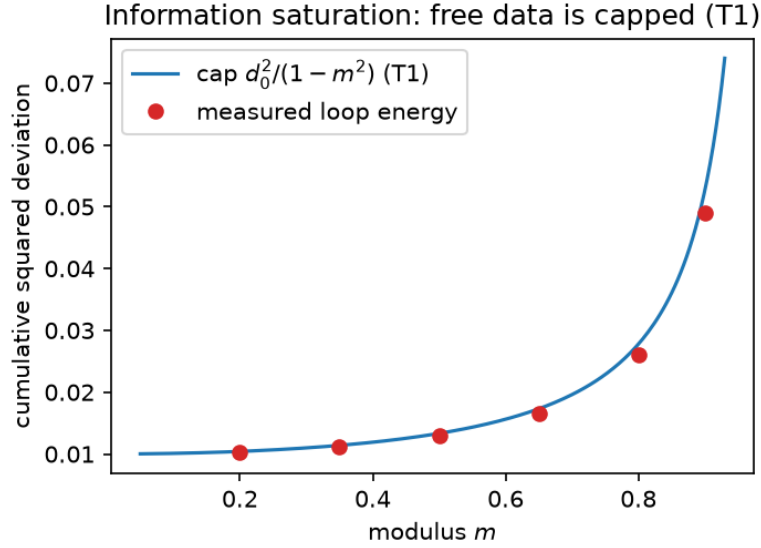


Figure S1: Information saturation (T1). Cumulative squared deviation of the noiseless retraining trajectory against the modulus m ; the cap $d_0^2/(1-m^2)$ (line) matches the measured loop energy (points). Without deliberate exploration, the information a retraining loop reveals about its own performativity is bounded, and the bound grows with m . Pivot E1 added to this picture the excitation floor $(\sigma/\gamma)^2/(1-m^2)$ that noisy observations contribute (Table S1, feedback-floor row).

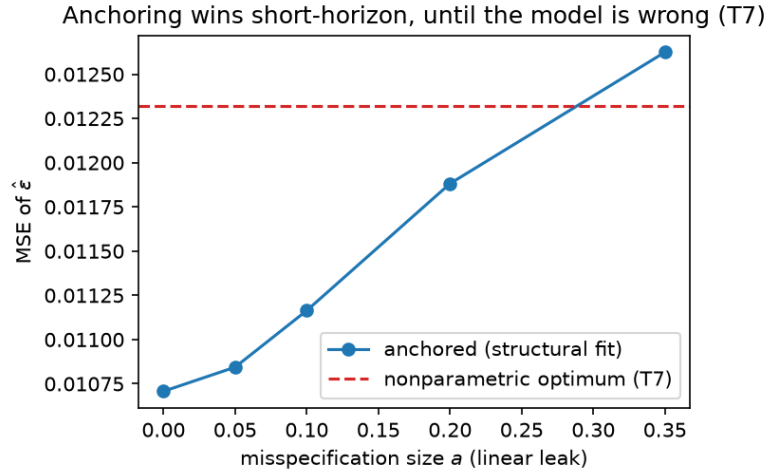


Figure S2: Anchoring crossover (T7). Mean squared error of the anchored structural fit against the size a of a linear misspecification the family cannot represent, compared with the budget-and-horizon-optimal nonparametric estimator (dashed). Anchoring wins at small misspecification and loses past the crossover near $a \approx 0.29$, as T7 predicts. Pivot E5 replaced an exponential misspecification, which had decayed to about 10^{-3} by the operating point and was correctly invisible to the fit, with this linear leak.

Certification before exploitation: the gate timeline (shaded = frozen)

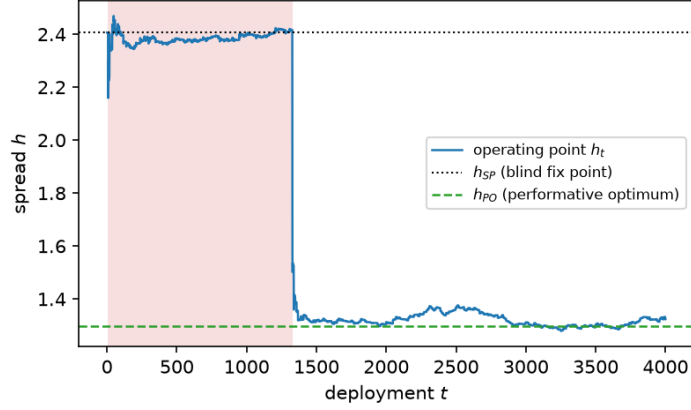


Figure S3: Certification before exploitation (L4, P7.1). Operating point h_t of the certified agent over 4,000 deployments. In the shaded interval the confidence sequence cannot yet certify that correcting is safe, so the agent stays frozen near the blind fixed point h_{SP} ; once certified, it moves to the performative optimum h_{PO} and remains inside the trust region (Table S1: 0 violations).

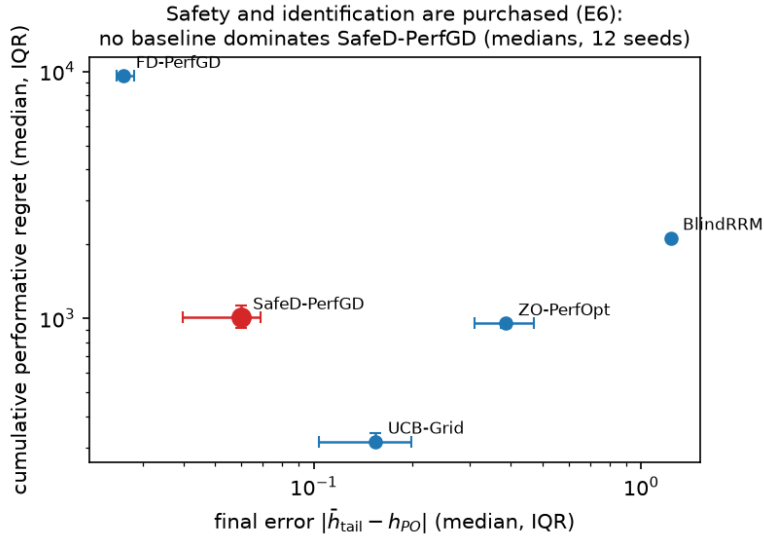


Figure S4: Baseline comparison (E6). Median final error $|\bar{h}_{tail} - h_{PO}|$ against median cumulative performative regret over 12 seeds (bars: interquartile range) for the certified agent (SafeD-PerfGD) and four baselines. No baseline dominates it: FD-PerfGD attains lower error at roughly ten times the regret, and UCB-Grid lower regret at higher error. Safety and identification are purchased, not free.

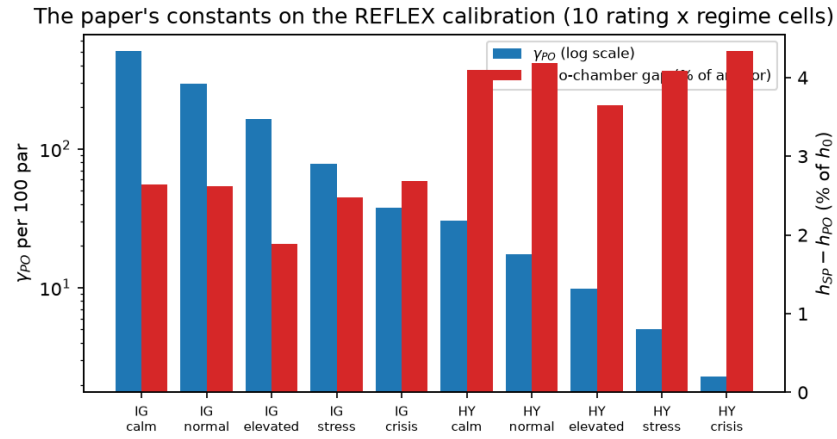


Figure S5: Real-data calibration (Table S1, calibration row). The curvature γ_{PO} (log scale, left axis) and the echo-chamber gap $h_{SP} - h_{PO}$ as a percentage of the anchor (right axis) across the ten rating-by-regime cells of the REFLEX corporate-bond calibration [Nagarajan and Ashok, 2026] (IG: investment grade; HY: high yield). All ten cells reproduce within 0.8 percent.

E Reproducibility

What ships. The supplementary repository contains the derivation documents and the theorem register (26 results, 38 numerical derivation checks), the simulation pipeline (environments, estimators, designs, agents, baselines; 36 measured-versus-predicted rows; 9 unit tests), the real-data port with its 10-cell validation, and the eleven-pivot record in the pipeline README. Both verification suites run in continuous integration on every push in the project repository; the two workflow files are not part of this anonymized copy. The structural market used throughout, and the corporate-bond calibration behind the real-data row of Table S1, are those of REFLEX [Nagarajan and Ashok, 2026], on which this work builds. All results are deterministic from seeds, `numpy`-only, and CPU-only. Reported final errors are 300-step tail averages over the deployed path, stable within 0.01 under horizon perturbation, with a worst measured spread of 0.0062.

What is stated here but not proved here. This version proves Theorems 1 and 2 of the main text, the two results the main text argues from. The minimax floors and the design and certification results (T3, T4, T5a, C5.1, T6, T7, L4, P7.1) are proved in the supplementary repository, in `mlxor-derivations/latex/proofs.tex` and the corresponding derivation documents, with the register recording each result’s proof status.

Provenance of the numbers. Every quantity in the main text and in Table S1 is copied from a committed artifact: `results/RESULTS.md` for the harness rows, `results/OPEN1.md` for the reach, the witness, and the search history, `results/REALDATA.md` for the calibration port, `results/STABILITY.md` for the tail-average stability, `results/full_run.log` for quantities not surfaced in the summary tables, and `THEOREMS.md` for the register. The runs behind them were executed and recorded on 1 September 2026. This version re-narrates them and does not re-run them, since none of its changes touch their inputs.